Four-wire-verified dynamic operating envelopes with GENCO

Where a grid foundation model helps on four-wire LV networks, where it fails, and how we adapted it

In one sentence. On a four-wire LV benchmark feeder with real Flemish smart-meter households, dynamic envelopes free several MWh of capacity per household per year where a static limit frees none. We made GENCO work on LV, put it inside an exact four-wire check, extended a GENCO-style model to explicit phase and neutral states, and benchmarked every GENCO role against strong classical methods. We report honestly where it adds value.

The application

Operators must host EVs, heat pumps and rooftop PV without reinforcing the grid. A **dynamic operating envelope** (DOE) gives each household a time-varying import/export limit.

- **Feeder:** ENWL1, 800 kVA, 200 single-phase customers, populated with Fluvius 2024 profiles.
- **Scenarios:** 200 forecast scenarios per interval.
- **Acceptance:** every issued envelope must pass an **exact four-wire OpenDSS check** of a frozen rule (≥98% of scenarios feasible).

Scenarios → proposal (exact / GENCO-screened / learned) → exact four-wire check → envelope

Impact: capacity a static limit cannot free

4.81 MWh
export envelope-hours / household / yr

2.50 MWh
import envelope-hours / household / yr

0
static year-round allowance

2-3×
balanced-model envelopes: four-wire unsafe

- 36 held-out days; intervals export 4.35–5.30 and import 2.11–2.87 MWh.
- Robust across 5 household placements. The headline placement is the most conservative.
- Idealised three-phase HP/EV connection lifts capacity 2–2.5×.

GENCO development: making it work on LV

Direct transfer to a 398-bus LV feeder failed. Stock GENCO collapsed and lost to predicting the training mean. We diagnosed and fixed three causes:

- **Impedance blindness:** the edge encoder's final LayerNorm cancels $|Y|$. Fixed with asinh admittance features.
- **Physics-loss collapse** at LV scales. Fixed by a linear power-flow prior with a zero-initialised correction head.
- **Ill-conditioned flows** from voltage differences. Fixed with Kirchhoff-current-law flows.

Result: voltage MAE **1.0e-3 pu**. We also found a graphkit best-model export bug and a datakit 1.5 pu slack bug.

Four-wire GENCO-style extension (exploratory)

A GNN on the 397-bus **four-wire** graph (phases a, b, c and neutral, full 4×4 impedances) corrects a one-step four-wire sweep prior. On 2,968 held-out states:

−91%
phase-to-neutral voltage error

35.8%→5.8%
states sent to OpenDSS

There were zero observed false-safe decisions under the exact gate. Balanced models send 63% to OpenDSS.

Benchmarks: every GENCO role vs strong baselines

Screening (C/C++)	GENCO + exact gate matches the exact search on 100% of intervals, 6% fewer exact calls. Classical screens do as well.
Learned proposals	79% of exact capacity with **6.7% of the exact calls**, but phase features alone do the same. GENCO arm: −0.06 pp.
GENCO embeddings	Frozen balanced-GENCO embeddings made the four-wire model **worse** (+6.0 pp fallbacks).
Classical physics	An iterated four-wire sweep is 80× more accurate than any learned model. It remains the accuracy leader.

What we do not claim

- Unconditional safety. Envelopes pass a finite design-sample rule; only 47% pass a 95% lower-bound test on fresh draws.
- A GENCO advantage over specialised classical solvers, or a wall-clock speed-up. We report exact-call reductions.
- Generalisation beyond one benchmark feeder.

Open science and rigour

- Pre-registered core evaluation. Every other study was frozen before its results were seen, with all deviations disclosed as amendments.
- Incidents found and disclosed: wrong voltage base, OpenDSS engine poisoning, stale cluster caches.
- 72 automated tests. `reproduce_check.py` regenerates every headline result byte for byte. Run manifests hash every artifact.

Potential: one model across LV, MV and HV for TSO–DSO coordination; LV envelopes sum to the flexibility a DSO offers the TSO.

Four-wire extension: a GENCO-style GNN corrects a cheap four-wire prior; GENCO embeddings do not help

2,968 held-out states; zero observed false-safe decisions for every method. Blue = learned model that helps; orange = GENCO variants that do not; grey = physics baselines

